

NIST SP 800-53 Rev 5 control PE-3.3 (Continuous Guards) is defined as follows: PE-3.3 · Continuous Guards Control Statement: Employ guards to control [organization-defined parameter] to the facility where the system resides 24 hours per day, 7 days per week. Supplemental Guidance: Employ guards to control [organization-defined parameter] to the facility where the system resides 24 hours per day, 7 days per week. Employing guards at selected physical access points to the facility provides a more rapid response capability for organizations. Guards also provide the opportunity for human surveillance in areas of the facility not covered by video surveillance.

FedRAMP High Baseline · PE-14.2 · Monitoring with Alarms and Notifications Control Statement: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. Supplemental Guidance: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. The alarm or notification may be an audible alarm or a visual message in real time to personnel or roles defined by the organization. Such alarms and notifications can help minimize harm to individuals and damage to organizational assets by facilitating a timely incident response. Assessment Objectives: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. FedRAMP Parameter Values: pe-14.02_odp: personnel or roles to be notified by environmental control monitoring when environmental changes are potentially harmful to personnel or equipment is/are defined;

Under the FedRAMP High baseline, PE-6.1 (PE-6.1 Intrusion Alarms and Surveillance Equipment): FedRAMP High Baseline · PE-6.1 · Intrusion Alarms and Surveillance Equipment Control Statement: Monitor physical access to the facility where the system resides using physical intrusion alarms and surveillance equipment. Supplemental Guidance: Monitor physical access to the facility where the system resides using physical intrusion alarms and surveillance equipment. Physical intrusion alarms can be employed to alert security personnel when unauthorized access to the facility is attempted. Alar

Under the FedRAMP High baseline, PE-12 (PE-12 Emergency Lighting): FedRAMP High Baseline · PE-12 · Emergency Lighting Control Statement: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. Supplemental Guidance: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. The pro

PE-3.7 · Physical Barriers Control Statement: Limit access using physical barriers. Supplemental Guidance: Limit access using physical barriers. Physical barriers include bollards, concrete slabs, jersey walls, and hydraulic active vehicle barriers.

PE-9.2 · Automatic Voltage Controls Control Statement: Employ automatic voltage controls for [organization-defined parameter]. Supplemental Guidance: Employ automatic voltage controls for [organization-defined parameter]. Automatic voltage controls can monitor and control voltage. Such controls include voltage regulators, voltage conditioners, and voltage stabilizers.

NIST SP 800-171 Rev 3 control 03.10.02 (03.10.02 Monitoring Physical Access) requires organizations to implement the following: Control Statement: SR-03.10.02.a Monitor physical access to the facility where the system resides to detect and respond to physical security incidents. SR-03.10.02.b Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter]. For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures, and technical controls address each sub-requirement of 03.10.02.

Implementing 03.10.01 (03.10.01 Physical Access Authorizations) differs between cloud and on-premises environments in several ways: ****On-premises****: The organization has direct control over all infrastructure components. Implementation of 03.10.01 involves configuring local systems, network devices, and security tools. The organization is fully responsible for all aspects of the control. ****Cloud (IaaS/PaaS/SaaS)****: Responsibility is shared between the organization and the Cloud Service Provider (CSP). Under the FedRAMP shared responsibility model: - The CSP handles infrastructure-level controls (physical security, hypervisor, network backbone) - The organization

handles application-level controls, user access, and data protection - The split depends on the service model (IaaS gives more control, SaaS gives less) For CMMC Level 2, the organization must document in their SSP which components of 03.10.01 are implemented by the CSP (with reference to the CSP's FedRAMP authorization) and which are implemented by the organization. The CUI boundary must be clearly defined regardless of environment.

PE-18 · Location of System Components Control Statement: Position system components within the facility to minimize potential damage from [organization-defined parameter] and to minimize the opportunity for unauthorized access. Supplemental Guidance: Position system components within the facility to minimize potential damage from [organization-defined parameter] and to minimize the opportunity for unauthorized access. Physical and environmental hazards include floods, fires, tornadoes, earthquakes, hurricanes, terrorism, vandalism, an electromagnetic pulse, electrical interference, and other forms of incoming electromagnetic radiation. Organizations consider the location of entry points where unauthorized individuals, while not being granted access, might nonetheless be near systems. Such proximity can increase the risk of unauthorized access to organizational communications using wireless packet sniffers or microphones, or unauthorized disclosure of information.

NIST SP 800-53 Rev 5 control PE-14.2 (Monitoring with Alarms and Notifications) summarizes the requirements for this control: PE-14.2 · Monitoring with Alarms and Notifications Control Statement: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. Supplemental Guidance: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. The alarm or notification may be an audible alarm or a visu

NIST SP 800-53 Rev 5 control PE-6 (Monitoring Physical Access) is assessed by examining, interviewing, and testing: PE-6 · Monitoring Physical Access Control Statement: a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents; b. Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter] ; and c. Coordinate results of reviews and investigations with the organizational incident response capability. Supplemental Guidance: a. Monitor physical access to the facility where the system resides

NIST SP 800-171 Rev 3 control 03.10.07 (03.10.07 Physical Access Control) requires organizations to implement the following: Control Statement: SR-03.10.07.a Enforce physical access authorizations at entry and exit points to the facility where the system resides by: SR-03.10.07.a.01 Verifying individual physical access authorizations before granting access to the facility and SR-03.10.07.a.02 Controlling ingress and egress with physical access control systems, devices, or guards. SR-03.10.07.b Maintain physical access audit logs for entry or exit points. SR-03.10.07.c Escort visitors, and control visitor activity. SR-03.10.07.d Secure keys, combinations, and other physical access devices. SR-03.10.07.e Control physical access to output devices to prevent unauthorized individuals from obtaining access to CUI. For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures, and technical controls address each sub-requirement of 03.10.07.

NIST SP 800-53 Rev 5 control PE-9.2 (Automatic Voltage Controls) summarizes the requirements for this control: PE-9.2 · Automatic Voltage Controls Control Statement: Employ automatic voltage controls for [organization-defined parameter]. Supplemental Guidance: Employ automatic voltage controls for [organization-defined parameter]. Automatic voltage controls can monitor and control voltage. Such controls include voltage regulators, voltage conditioners, and voltage stabilizers.

NIST SP 800-53 Rev 5 control PE-3.4 (Lockable Casings) is defined as follows: PE-3.4 · Lockable Casings Control Statement: Use lockable physical casings to protect [organization-defined parameter] from unauthorized physical access. Supplemental Guidance: Use lockable physical casings to protect [organization-defined parameter] from unauthorized physical access. The greatest risk from the use of portable devices-such as smart phones, tablets, and notebook computers-is theft. Organizations can employ lockable, physical casings to reduce or eliminate the risk of equipment theft. Such casings come in a variety of sizes, from units that protect a single notebook computer t

PE-15 · Water Damage Protection Control Statement: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. Supplemental Guidance: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. The provision of water damage protection primarily applies to organizational facilities that contain concentrations of system resources, including data centers, server rooms, and mainframe computer rooms. Isolation valves can be employed in addition to or in lieu of master shutoff valves to shut off water supplies in specific areas of concern without affecting entire organizations.

NIST SP 800-171 Rev 3 control 03.10.06 (03.10.06 Alternate Work Site) falls under the Physical Protection family. Control Statement: SR-03.10.06.a Determine alternate work sites allowed for use by employees. SR-03.10.06.b Employ the following security requirements at alternate work sites: [organization-defined parameter]. Supplemental Guidance: Alternate work sites include the private residences of employees or other facilities designated by the organization. Alternate work sites can provide readily available alternate locations during contingency operations. Organizations can define different security requirements for specific alternate work sites or types of sites, depending on the work-related activities conducted at the sites. Assessing the effectiveness of the requirements and providing a means to communicate

NIST SP 800-53 Rev 5 control PE-23 (Facility Location) is assessed by examining, interviewing, and testing: PE-23 · Facility Location Control Statement: a. Plan the location or site of the facility where the system resides considering physical and environmental hazards; and b. For existing facilities, consider the physical and environmental hazards in the organizational risk management strategy. Supplemental Guidance: a. Plan the location or site of the facility where the system resides considering physical and environmental hazards; and b. For existing facilities, consider the physical and environment

NIST SP 800-53 Rev 5 control PE-6.2 (Automated Intrusion Recognition and Responses) is defined as follows: PE-6.2 · Automated Intrusion Recognition and Responses Control Statement: Recognize [organization-defined parameter] and initiate [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Recognize [organization-defined parameter] and initiate [organization-defined parameter] using [organization-defined parameter]. Response actions can include notifying selected organizational personnel or law enforcement personnel. Automated mechanisms implemented to initiate response actions include system alert notifications, email and text messages, and activating do

NIST SP 800-53 Rev 5 control PE-9 (Power Equipment and Cabling) summarizes the requirements for this control: PE-9 · Power Equipment and Cabling Control Statement: Protect power equipment and power cabling for the system from damage and destruction. Supplemental Guidance: Protect power equipment and power cabling for the system from damage and destruction. Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation. Types of power equipment and

Under the FedRAMP High baseline, PE-9 (PE-9 Power Equipment and Cabling): FedRAMP High Baseline · PE-9 · Power Equipment and Cabling Control Statement: Protect power equipment and power cabling for the system from damage and destruction. Supplemental Guidance: Protect power equipment and power cabling for the system from damage and destruction. Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation. Typ

NIST SP 800-171 Rev 3 control 03.10.02 (03.10.02 Monitoring Physical Access) is assessed by examining the following: Assessment Objectives: - physical access to the facility where the system resides is monitored to detect physical security incidents. - physical security incidents are responded to. - physical access logs are reviewed [organization-defined parameter] . - physical access logs are reviewed upon occurrence of [organization-defined parameter].

From the DoD SP 800-171 Rev 3 Organization-Defined Parameters: Related Controls: PE-02 ODP Values ODP Identifier ODP Assignment Text ODP Value 03.10.01.c [Assignment: organization-defined

frequency] at least every 12 months, or when there are significant incidents or significant changes to risks. Physical Protection 3.10.2 Physical Access Monitoring and Review a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents. b. Review physical access logs [Assignment: organization-defined frequency] (03.10.02.b.01) and upon occurrence of [Assignment: organization-defined events or potential indications of events] (03.10.02.b.02). Related Controls: PE-06 ODP Values ODP Identifier ODP Assignment Text ODP Value 03.10.02.b.01 [Assignment: organization-defined frequency] at least

Control PE-5: ACCESS CONTROL FOR OUTPUT DEVICES Control: Control physical access to output from [Assignment: organization-defined output devices] to prevent unauthorized individuals from obtaining the output. Discussion: Controlling physical access to output devices includes placing output devices in locked rooms or other secured areas with keypad or card reader access controls and allowing access to authorized individuals only, placing output devices in locations that can be monitored by personnel, installing monitor or screen filters, and using headphones. Examples of output devices include monitors, printers, scanners, audio devices, facsimile machines, and copiers. Related Controls: PE-2, PE-3, PE-4, PE-18. Control Enhancements: (1) ACCESS CONTROL FOR OUTPUT DEVICES | ACCESS TO OUTPUT BY AUTHORIZED INDIVIDUALS [Withdrawn: Incorporated into PE-5.] (2) ACCESS CONTROL FOR OUTPUT DEVICES | LINK TO INDIVIDUAL IDENTITY Link individual identity to receipt of output from output devices. Discussion: Methods for linking individual identity to the receipt of output from output devices include installing security functionality on facsimile machines, copiers, and printers. Such functionality allows organizations to implement authentication on output devices prior to the release of output to individuals. CHAPTER THREE PAGE 183 This publication is available free of charge from: <https://doi.org/10.6028/NIST.SP.800-53r5>. NIST SP 800-53, REV. 5 SECURITY AND PRIVACY CONTROLS FOR INFORMATION SYSTEMS AND ORGANIZATIONS

Related Controls: None. (3) ACCESS CONTROL FOR OUTPUT DEVICES | MARKING OUTPUT DEVICES [Withdrawn: Incorporated into PE-22.] References: [IR 8023].

NIST SP 800-53 Rev 5 control PE-17 (Alternate Work Site) is assessed by examining, interviewing, and testing: PE-17 · Alternate Work Site Control Statement: a. Determine and document the [organization-defined parameter] allowed for use by employees; b. Employ the following controls at alternate work sites: [organization-defined parameter]; c. Assess the effectiveness of controls at alternate work sites; and d. Provide a means for employees to communicate with information security and privacy personnel in case of incidents. Supplemental Guidance: a. Determine and document the [organization-defined para

NIST SP 800-53 Rev 5 control PE-6 (Monitoring Physical Access) summarizes the requirements for this control: PE-6 · Monitoring Physical Access Control Statement: a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents; b. Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter] ; and c. Coordinate results of reviews and investigations with the organizational incident response capability. Supplemental Guidance: a. Monitor physical access to the facility where the system resides

NIST SP 800-53 Rev 5 control PE-16 (Delivery and Removal) is assessed by examining, interviewing, and testing: PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas and isolating the areas fro

NIST SP 800-53 Rev 5 control PE-19 (Information Leakage) is assessed by examining, interviewing, and testing: PE-19 · Information Leakage Control Statement: Protect the system from information leakage due to electromagnetic signals emanations. Supplemental Guidance: Protect the system from information leakage due to electromagnetic signals emanations. Information leakage is the intentional or unintentional release of data or information to an untrusted environment from electromagnetic signals emanations. The security categories or classifications of systems (with respect to confidentiality), organizat

NIST SP 800-53 Rev 5 control PE-19.1 (National Emissions Policies and Procedures) is defined as follows: PE-19.1 · National Emissions Policies and Procedures Control Statement: Protect system components, associated data communications, and networks in accordance with national Emissions Security policies and procedures based on the security category or classification of the information. Supplemental Guidance: Protect system components, associated data communications, and networks in accordance with national Emissions Security policies and procedures based on the security category or classification of the information. Emissions Security (EMSEC) policies include the former TEMPEST policies.

NIST SP 800-53 Rev 5 control PE-15.1 (Automation Support) is assessed by examining, interviewing, and testing: PE-15.1 · Automation Support Control Statement: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Automated mechanisms include notification systems, water detection sensors, and alarms.

NIST SP 800-53 Rev 5 control PE-9.2 (Automatic Voltage Controls) is assessed by examining, interviewing, and testing: PE-9.2 · Automatic Voltage Controls Control Statement: Employ automatic voltage controls for [organization-defined parameter]. Supplemental Guidance: Employ automatic voltage controls for [organization-defined parameter]. Automatic voltage controls can monitor and control voltage. Such controls include voltage regulators, voltage conditioners, and voltage stabilizers.

NIST SP 800-53 Rev 5 control PE-3.8 (Access Control Vestibules) summarizes the requirements for this control: PE-3.8 · Access Control Vestibules Control Statement: Employ access control vestibules at [organization-defined parameter]. Supplemental Guidance: Employ access control vestibules at [organization-defined parameter]. An access control vestibule is part of a physical access control system that typically provides a space between two sets of interlocking doors. Vestibules are designed to prevent unauthorized individuals from following authorized individuals into facilities with controlled access.

NIST SP 800-53 Rev 5 control PE-3.8 (Access Control Vestibules) is assessed by examining, interviewing, and testing: PE-3.8 · Access Control Vestibules Control Statement: Employ access control vestibules at [organization-defined parameter]. Supplemental Guidance: Employ access control vestibules at [organization-defined parameter]. An access control vestibule is part of a physical access control system that typically provides a space between two sets of interlocking doors. Vestibules are designed to prevent unauthorized individuals from following authorized individuals into facilities with controlled access.

Implementing 03.10.06 (03.10.06 Alternate Work Site) differs between cloud and on-premises environments in several ways: ****On-premises****: The organization has direct control over all infrastructure components. Implementation of 03.10.06 involves configuring local systems, network devices, and security tools. The organization is fully responsible for all aspects of the control. ****Cloud (IaaS/PaaS/SaaS)****: Responsibility is shared between the organization and the Cloud Service Provider (CSP). Under the FedRAMP shared responsibility model: - The CSP handles infrastructure-level controls (physical security, hypervisor, network backbone) - The organization handles application-level controls, user access, and data protection - The split depends on the service model (IaaS gives more control, SaaS gives less) For CMMC Level 2, the organization must document in their SSP which components of 03.10.06 are implemented by the CSP (with reference to the CSP's FedRAMP authorization) and which are implemented by the organization. The CUI boundary must be clearly defined regardless of environment.

Under the FedRAMP Moderate baseline, PE-13 (PE-13 Fire Protection): FedRAMP Moderate Baseline · PE-13 · Fire Protection Control Statement: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. Supplemental Guidance: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. The provision of fire detection and suppression systems applies primarily to organizational facilities that contain concentrations of system resources, including data centers, server

PE-2 · Physical Access Authorizations Control Statement: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access list when access is no longer required. Supplemental Guidance: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access list when access is no longer required. Physical access authorizations apply to employees and visitors. Individuals with permanent physical access authorization credentials are not considered visitors. Authorization credentials include ID badges, identification cards, and smart cards. Organizations determine the strength of authorization credentials needed consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Physical access authorizations may not be necessary to access certain areas within facilities that are designated as publicly accessible.

NIST SP 800-171 Rev 3 control 03.10.01 (03.10.01 Physical Access Authorizations) maps to corresponding controls in NIST SP 800-53 Rev 5 under the Physical Protection (PE) family. The CMMC Level 2 assessment guide provides the specific mapping. Organizations implementing both frameworks should use the NIST SP 800-171 Rev 3 to SP 800-53 Rev 5 crosswalk to identify overlapping controls and avoid duplicate effort.

NIST SP 800-53 Rev 5 control PE-16 (Delivery and Removal): PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and ex

Under the FedRAMP High baseline, PE-15.1 (PE-15.1 Automation Support): FedRAMP High Baseline · PE-15.1 · Automation Support Control Statement: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Automated mechanisms include notification systems, water detection sensors, and alarms. Assessment Objectives: Detect the presence of water near t

NIST SP 800-171 Rev 3 control 03.10.01 (03.10.01 Physical Access Authorizations) is assessed by examining the following: Assessment Objectives: - a list of individuals with authorized access to the facility where the system resides is developed. - a list of individuals with authorized access to the facility where the system resides is approved. - a list of individuals with authorized access to the facility where the system resides is maintained. - the facility access list is reviewed [organization-defined parameter]. - individuals from the facility access list are removed when access is no longer required. - authorization credentials for facility access are issued.

NIST SP 800-53 Rev 5 control PE-22 (Component Marking) is defined as follows: PE-22 · Component Marking Control Statement: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Supplemental Guidance: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Hardware components that may require marking include input and output devices. Input devices include desktop and notebook computers, keyboards, tablets, and s

NIST SP 800-53 Rev 5 control PE-6.2 (Automated Intrusion Recognition and Responses) is assessed by examining, interviewing, and testing: PE-6.2 · Automated Intrusion Recognition and Responses Control Statement: Recognize [organization-defined parameter] and initiate [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Recognize [organization-defined parameter] and initiate [organization-defined parameter] using [organization-defined parameter]. Response actions can include notifying selected organizational personnel or law enforcement personnel. Automated mechanisms implemented to init

NIST SP 800-53 Rev 5 control PE-9 (Power Equipment and Cabling) is assessed by examining,

interviewing, and testing: PE-9 · Power Equipment and Cabling Control Statement: Protect power equipment and power cabling for the system from damage and destruction. Supplemental Guidance: Protect power equipment and power cabling for the system from damage and destruction. Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation. Types of power equipment an

NIST SP 800-53 Rev 5 control PE-3 (Physical Access Control) summarizes the requirements for this control: PE-3 · Physical Access Control Control Statement: a. Enforce physical access authorizations at [organization-defined parameter] by: 1. Verifying individual access authorizations before granting access to the facility; and 2. Controlling ingress and egress to the facility using [organization-defined parameter]; b. Maintain physical access audit logs for [organization-defined parameter]; c. Control access to areas within the facility designated as publicly accessible by implementing the follo

NIST SP 800-53 Rev 5 control PE-8.1 (Automated Records Maintenance and Review) is defined as follows: PE-8.1 · Automated Records Maintenance and Review Control Statement: Maintain and review visitor access records using [organization-defined parameter]. Supplemental Guidance: Maintain and review visitor access records using [organization-defined parameter]. Visitor access records may be stored and maintained in a database management system that is accessible by organizational personnel. Automated access to such records facilitates record reviews on a regular basis to determine if access authorizations are current and still required to support organizational mission and business functions.

Under the FedRAMP High baseline, PE-13.1 (PE-13.1 Detection Systems · Automatic Activation and Notification): FedRAMP High Baseline · PE-13.1 · Detection Systems · Automatic Activation and Notification Control Statement: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Supplemental Guidance: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Organizations can identify personnel,

NIST SP 800-171 Rev 3 control 03.10.01 (03.10.01 Physical Access Authorizations) is assessed by examining the following: Assessment Objectives: - a list of individuals with authorized access to the facility where the system resides is developed. - a list of individuals with authorized access to the facility where the system resides is approved. - a list of individuals with authorized access to the facility where the system resides is maintained. - the facility access list is reviewed [organization-defined parameter]. - individuals from the facility access list are removed when access is no longer required. - authorization credentials for facility access are issued.

NIST SP 800-171 Rev 3 control 03.10.06 (03.10.06 Alternate Work Site) is assessed by examining the following: Assessment Objectives: - alternate work sites allowed for use by employees are determined. - the following security requirements are employed at alternate work sites: [organization-defined parameter].

PE-22 · Component Marking Control Statement: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Supplemental Guidance: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Hardware components that may require marking include input and output devices. Input devices include desktop and notebook computers, keyboards, tablets, and smart phones. Output devices include printers, monitors/video displays, facsimile machines, scanners, copiers, and audio devices. Permissions controlling output to the output devices are addressed in [AC-3](#ac-3) or [AC-4](#ac-4) . Components are marked to indicate the impact level or classification level of the system to which the devices are connected, or the impact level or classification level of the information permitted to be output. Security marking refers to the use of human-readable security attributes. Security labeling refers to the use of security attributes for internal system data structures. Security marking is generally not required for hardware components that process, store, or transmit information determined by organizations to be in the public domain or to be publicly releasable. However, organizations may require markings for hardware

components that process, store, or transmit public information in order to indicate that such information is publicly releasable. Marking of system hardware components reflects applicable laws, executive orders, directives, policies, regulations, and standards.

Under the FedRAMP High baseline, PE-5 (PE-5 Access Control for Output Devices): FedRAMP High Baseline · PE-5 · Access Control for Output Devices Control Statement: Control physical access to output from [organization-defined parameter] to prevent unauthorized individuals from obtaining the output. Supplemental Guidance: Control physical access to output from [organization-defined parameter] to prevent unauthorized individuals from obtaining the output. Controlling physical access to output devices includes placing output devices in locked rooms or other secured areas with

Under the FedRAMP High baseline, PE-3 (PE-3 Physical Access Control): FedRAMP High Baseline · PE-3 · Physical Access Control Control Statement: a. Enforce physical access authorizations at [organization-defined parameter] by: 1. Verifying individual access authorizations before granting access to the facility; and 2. Controlling ingress and egress to the facility using [organization-defined parameter]; b. Maintain physical access audit logs for [organization-defined parameter]; c. Control access to areas within the facility designated as publicly accessible b

NIST SP 800-171 Rev 3 control 03.10.02 (03.10.02 Monitoring Physical Access) maps to corresponding controls in NIST SP 800-53 Rev 5 under the Physical Protection (PE) family. The CMMC Level 2 assessment guide provides the specific mapping. Organizations implementing both frameworks should use the NIST SP 800-171 Rev 3 to SP 800-53 Rev 5 crosswalk to identify overlapping controls and avoid duplicate effort.

Under the FedRAMP High baseline, PE-13 (PE-13 Fire Protection): FedRAMP High Baseline · PE-13 · Fire Protection Control Statement: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. Supplemental Guidance: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. The provision of fire detection and suppression systems applies primarily to organizational facilities that contain concentrations of system resources, including data centers, server room

NIST SP 800-53 Rev 5 control PE-6.4 (Monitoring Physical Access to Systems) is assessed by examining, interviewing, and testing: PE-6.4 · Monitoring Physical Access to Systems Control Statement: Monitor physical access to the system in addition to the physical access monitoring of the facility at [organization-defined parameter]. Supplemental Guidance: Monitor physical access to the system in addition to the physical access monitoring of the facility at [organization-defined parameter]. Monitoring physical access to systems provides additional monitoring for those areas within facilities where there is a concentration o

NIST SP 800-53 Rev 5 control PE-12 (Emergency Lighting) is assessed by examining, interviewing, and testing: PE-12 · Emergency Lighting Control Statement: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. Supplemental Guidance: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. The provision of emergency ligh

NIST SP 800-53 Rev 5 control PE-3.3 (Continuous Guards) is assessed by examining, interviewing, and testing: PE-3.3 · Continuous Guards Control Statement: Employ guards to control [organization-defined parameter] to the facility where the system resides 24 hours per day, 7 days per week. Supplemental Guidance: Employ guards to control [organization-defined parameter] to the facility where the system resides 24 hours per day, 7 days per week. Employing guards at selected physical access points to the facility provides a more rapid response capability for organizations. Guards also provide the opportun

NIST SP 800-53 Rev 5 control PE-10 (Emergency Shutoff) is assessed by examining, interviewing, and testing: PE-10 · Emergency Shutoff Control Statement: a. Provide the capability of shutting off power to [organization-defined parameter] in emergency situations; b. Place emergency shutoff switches or devices in [organization-defined parameter] to facilitate access for authorized personnel; and c. Protect emergency power shutoff capability from unauthorized activation.

Supplemental Guidance: a. Provide the capability of shutting off power to [organization-defined parameter] in emergency situations; b.

NIST SP 800-53 Rev 5 control PE-13.1 (Detection Systems · Automatic Activation and Notification) summarizes the requirements for this control: PE-13.1 · Detection Systems · Automatic Activation and Notification Control Statement: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Supplemental Guidance: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Organizations can identify personnel, roles, and emergency res

NIST SP 800-53 Rev 5 control PE-3.5 (Tamper Protection) is defined as follows: PE-3.5 · Tamper Protection Control Statement: Employ [organization-defined parameter] to [organization-defined parameter] physical tampering or alteration of [organization-defined parameter] within the system. Supplemental Guidance: Employ [organization-defined parameter] to [organization-defined parameter] physical tampering or alteration of [organization-defined parameter] within the system. Organizations can implement tamper detection and prevention at selected hardware components or implement tamper detection at some components and tamper prevention at other components. Detection and pre

NIST SP 800-53 Rev 5 control PE-11 (Emergency Power) summarizes the requirements for this control: PE-11 · Emergency Power Control Statement: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. Supplemental Guidance: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. An uninterruptible power supply (UPS) is an electrical system or mechanism that provides emergency power when there is a failure of the main power source. A UPS is typi

Under the FedRAMP High baseline, PE-2 (PE-2 Physical Access Authorizations): FedRAMP High Baseline · PE-2 · Physical Access Authorizations Control Statement: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access list when access is no longer required. Supplemental Guidance: a. Develop, a

NIST SP 800-53 Rev 5 control PE-6.3 (Video Surveillance) summarizes the requirements for this control: PE-6.3 · Video Surveillance Control Statement: (a) Employ video surveillance of [organization-defined parameter]; (b) Review video recordings [organization-defined parameter] ; and (c) Retain video recordings for [organization-defined parameter]. Supplemental Guidance: (a) Employ video surveillance of [organization-defined parameter]; (b) Review video recordings [organization-defined parameter] ; and (c) Retain video recordings for [organization-defined parameter]. Video surveillance focuses o

Control PE-8: ds facilitates record reviews on a regular basis to determine if access authorizations are current and still required to support organizational mission and business functions. Related Controls: None. (2) VISITOR ACCESS RECORDS | PHYSICAL ACCESS RECORDS [Withdrawn: Incorporated into PE-2.] (3) VISITOR ACCESS RECORDS | LIMIT PERSONALLY IDENTIFIABLE INFORMATION ELEMENTS Limit personally identifiable information contained in visitor access records to the following elements identified in the privacy risk assessment: [Assignment: organization- defined elements]. Discussion: Organizations may have requirements that specify the contents of visitor access records. Limiting personally identifiable information in visitor access records when such information is not needed for operational purposes helps reduce the level of privacy risk created by a system. Related Controls: RA-3, SA-8. References: None.

PE-21 · Electromagnetic Pulse Protection Control Statement: Employ [organization-defined parameter] against electromagnetic pulse damage for [organization-defined parameter]. Supplemental Guidance: Employ [organization-defined parameter] against electromagnetic pulse damage for [organization-defined parameter]. An electromagnetic pulse (EMP) is a short burst of electromagnetic energy that is spread over a range of frequencies. Such energy bursts may be natural or man-made. EMP interference may be disruptive or damaging to electronic equipment. Protective measures used to

mitigate EMP risk include shielding, surge suppressors, ferro-resonant transformers, and earth grounding. EMP protection may be especially significant for systems and applications that are part of the U.S. critical infrastructure.

Control PE-15: WATER DAMAGE PROTECTION Control: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. Discussion: The provision of water damage protection primarily applies to organizational facilities that contain concentrations of system resources, including data centers, server rooms, and mainframe computer rooms. Isolation valves can be employed in addition to or in lieu of master shutoff valves to shut off water supplies in specific areas of concern without affecting entire organizations. Related Controls: AT-3, PE-10. Control Enhancements: (1) WATER DAMAGE PROTECTION | AUTOMATION SUPPORT Detect the presence of water near the system and alert [Assignment: organization-defined personnel or roles] using [Assignment: organization-defined automated mechanisms]. Discussion: Automated mechanisms include notification systems, water detection sensors, and alarms. Related Controls: None. References: None.

NIST SP 800-53 Rev 5 control PE-14 (Environmental Controls) summarizes the requirements for this control: PE-14 · Environmental Controls Control Statement: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organization-defined parameter]. Supplemental Guidance: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organization-defined parameter].

NIST SP 800-53 Rev 5 control PE-1 (Policy and Procedures) summarizes the requirements for this control: PE-1 · Policy and Procedures Control Statement: a. Develop, document, and disseminate to [organization-defined parameter]: 1. [organization-defined parameter] physical and environmental protection policy that: 2. Procedures to facilitate the implementation of the physical and environmental protection policy and the associated physical and environmental protection controls; b. Designate an [organization-defined parameter] to manage the development, documentation, and dissemination of the phy

NIST SP 800-53 Rev 5 control PE-2.3 (Restrict Unescorted Access) is assessed by examining, interviewing, and testing: PE-2.3 · Restrict Unescorted Access Control Statement: Restrict unescorted access to the facility where the system resides to personnel with [organization-defined parameter]. Supplemental Guidance: Restrict unescorted access to the facility where the system resides to personnel with [organization-defined parameter]. Individuals without required security clearances, access approvals, or need to know are escorted by individuals with appropriate physical access authorizations to ensure that infor

Under the FedRAMP Moderate baseline, PE-14 (PE-14 Environmental Controls): FedRAMP Moderate Baseline · PE-14 · Environmental Controls Control Statement: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organization-defined parameter]. Supplemental Guidance: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [orga

NIST SP 800-53 Rev 5 control PE-2 (Physical Access Authorizations) summarizes the requirements for this control: PE-2 · Physical Access Authorizations Control Statement: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access list when access is no longer required. Supplemental Guidance: a. Develop, approve, and maintain a l

NIST SP 800-171 Rev 3 control 03.10.06 (03.10.06 Alternate Work Site) requires organizations to implement the following: Control Statement: SR-03.10.06.a Determine alternate work sites allowed for use by employees. SR-03.10.06.b Employ the following security requirements at alternate work sites: [organization-defined parameter]. For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures,

and technical controls address each sub-requirement of 03.10.06.

NIST SP 800-53 Rev 5 control PE-3.4 (Lockable Casings) is assessed by examining, interviewing, and testing: PE-3.4 · Lockable Casings Control Statement: Use lockable physical casings to protect [organization-defined parameter] from unauthorized physical access. Supplemental Guidance: Use lockable physical casings to protect [organization-defined parameter] from unauthorized physical access. The greatest risk from the use of portable devices-such as smart phones, tablets, and notebook computers-is theft. Organizations can employ lockable, physical casings to reduce or eliminate the risk of equipment

NIST SP 800-53 Rev 5 control PE-8 (Visitor Access Records) is defined as follows: PE-8 · Visitor Access Records Control Statement: a. Maintain visitor access records to the facility where the system resides for [organization-defined parameter]; b. Review visitor access records [organization-defined parameter] ; and c. Report anomalies in visitor access records to [organization-defined parameter]. Supplemental Guidance: a. Maintain visitor access records to the facility where the system resides for [organization-defined parameter]; b. Review visitor access records [organization-defined parameter] ; and c. Report anomalies in visitor access records to [organization-defined

NIST SP 800-53 Rev 5 control PE-2.1 (Access by Position or Role) is assessed by examining, interviewing, and testing: PE-2.1 · Access by Position or Role Control Statement: Authorize physical access to the facility where the system resides based on position or role. Supplemental Guidance: Authorize physical access to the facility where the system resides based on position or role. Role-based facility access includes access by authorized permanent and regular/routine maintenance personnel, duty officers, and emergency medical staff.

FedRAMP High Baseline · PE-13.1 · Detection Systems · Automatic Activation and Notification Control Statement: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Supplemental Guidance: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. Organizations can identify personnel, roles, and emergency responders if individuals on the notification list need to have access authorizations or clearances (e.g., to enter to facilities where access is restricted due to the classification or impact level of information within the facility). Notification mechanisms may require independent energy sources to ensure that the notification capability is not adversely affected by the fire. Assessment Objectives: Employ fire detection systems that activate automatically and notify [organization-defined parameter] and [organization-defined parameter] in the event of a fire. FedRAMP Parameter Values: pe-13.01_odp.01: service provider building maintenance/physical security personnel pe-13.01_odp.02: service provider emergency responders with incident response responsibilities

Under the FedRAMP Low baseline, PE-13 (PE-13 Fire Protection): FedRAMP Low Baseline · PE-13 · Fire Protection Control Statement: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. Supplemental Guidance: Employ and maintain fire detection and suppression systems that are supported by an independent energy source. The provision of fire detection and suppression systems applies primarily to organizational facilities that contain concentrations of system resources, including data centers, server room

NIST SP 800-53 Rev 5 control PE-6.3 (Video Surveillance) is assessed by examining, interviewing, and testing: PE-6.3 · Video Surveillance Control Statement: (a) Employ video surveillance of [organization-defined parameter]; (b) Review video recordings [organization-defined parameter] ; and (c) Retain video recordings for [organization-defined parameter]. Supplemental Guidance: (a) Employ video surveillance of [organization-defined parameter]; (b) Review video recordings [organization-defined parameter] ; and (c) Retain video recordings for [organization-defined parameter]. Video surveillance focuses o

NIST SP 800-53 Rev 5 control PE-5 (Access Control for Output Devices) is assessed by examining, interviewing, and testing: PE-5 · Access Control for Output Devices Control Statement: Control physical access to output from [organization-defined parameter] to prevent unauthorized individuals from obtaining the output. Supplemental Guidance: Control physical access to output from [organization-defined parameter] to prevent unauthorized individuals from obtaining the output.

Controlling physical access to output devices includes placing output devices in locked rooms or other secured areas with keypad or card reader a

NIST SP 800-53 Rev 5 control PE-21 (Electromagnetic Pulse Protection) summarizes the requirements for this control: PE-21 · Electromagnetic Pulse Protection Control Statement: Employ [organization-defined parameter] against electromagnetic pulse damage for [organization-defined parameter]. Supplemental Guidance: Employ [organization-defined parameter] against electromagnetic pulse damage for [organization-defined parameter]. An electromagnetic pulse (EMP) is a short burst of electromagnetic energy that is spread over a range of frequencies. Such energy bursts may be natural or man-made. EMP interference may

Under the FedRAMP Moderate baseline, PE-4 (PE-4 Access Control for Transmission): FedRAMP Moderate Baseline · PE-4 · Access Control for Transmission Control Statement: Control physical access to [organization-defined parameter] within organizational facilities using [organization-defined parameter]. Supplemental Guidance: Control physical access to [organization-defined parameter] within organizational facilities using [organization-defined parameter]. Security controls applied to system distribution and transmission lines prevent accidental damage, disruption, and physical

Under the FedRAMP High baseline, PE-3.1 (PE-3.1 System Access): FedRAMP High Baseline · PE-3.1 · System Access Control Statement: Enforce physical access authorizations to the system in addition to the physical access controls for the facility at [organization-defined parameter]. Supplemental Guidance: Enforce physical access authorizations to the system in addition to the physical access controls for the facility at [organization-defined parameter]. Control of physical access to the system provides additional physical security for those areas within facil

NIST SP 800-53 Rev 5 control PE-22 (Component Marking) is assessed by examining, interviewing, and testing: PE-22 · Component Marking Control Statement: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Supplemental Guidance: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Hardware components that may require marking include input

Under the FedRAMP Moderate baseline, PE-11 (PE-11 Emergency Power): FedRAMP Moderate Baseline · PE-11 · Emergency Power Control Statement: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. Supplemental Guidance: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. An uninterruptible power supply (UPS) is an electrical system or mechanism that provides emergency power when there is a failure of the main

NIST SP 800-53 Rev 5 control PE-22 (Component Marking) summarizes the requirements for this control: PE-22 · Component Marking Control Statement: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Supplemental Guidance: Mark [organization-defined parameter] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component. Hardware components that may require marking include input

NIST SP 800-171 Rev 3 control 03.10.07 (03.10.07 Physical Access Control) is assessed by examining the following: Assessment Objectives: - physical access authorizations are enforced at entry and exit points to the facility where the system resides by verifying individual physical access authorizations before granting access. - physical access authorizations are enforced at entry and exit points to the facility where the system resides by controlling ingress and egress with physical access control systems, devices, or guards. - physical access audit logs for entry or exit points are maintained. - visitors are escorted. - visitor activity is controlled. - physical access to output devices is controlled to

NIST SP 800-171 Rev 3 control 03.10.06 (03.10.06 Alternate Work Site) is assessed by examining the following: Assessment Objectives: - alternate work sites allowed for use by employees are determined. - the following security requirements are employed at alternate work sites:

[organization-defined parameter].

PE-12.1 · Essential Mission and Business Functions Control Statement: Provide emergency lighting for all areas within the facility supporting essential mission and business functions. Supplemental Guidance: Provide emergency lighting for all areas within the facility supporting essential mission and business functions. Organizations define their essential missions and functions.

Under the FedRAMP Moderate baseline, PE-12 (PE-12 Emergency Lighting): FedRAMP Moderate Baseline · PE-12 · Emergency Lighting Control Statement: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. Supplemental Guidance: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. The

NIST SP 800-53 Rev 5 control PE-12.1 (Essential Mission and Business Functions) is defined as follows: PE-12.1 · Essential Mission and Business Functions Control Statement: Provide emergency lighting for all areas within the facility supporting essential mission and business functions. Supplemental Guidance: Provide emergency lighting for all areas within the facility supporting essential mission and business functions. Organizations define their essential missions and functions.

Under the FedRAMP Low baseline, PE-14 (PE-14 Environmental Controls): FedRAMP Low Baseline · PE-14 · Environmental Controls Control Statement: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organization-defined parameter]. Supplemental Guidance: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organizat

PE-14.2 · Monitoring with Alarms and Notifications Control Statement: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. Supplemental Guidance: Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [organization-defined parameter]. The alarm or notification may be an audible alarm or a visual message in real time to personnel or roles defined by the organization. Such alarms and notifications can help minimize harm to individuals and damage to organizational assets by facilitating a timely incident response.

Under the FedRAMP High baseline, PE-14 (PE-14 Environmental Controls): FedRAMP High Baseline · PE-14 · Environmental Controls Control Statement: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organization-defined parameter]. Supplemental Guidance: a. Maintain [organization-defined parameter] levels within the facility where the system resides at [organization-defined parameter] ; and b. Monitor environmental control levels [organiza

From the DoD SP 800-171 Rev 3 Organization-Defined Parameters: Physical Protection 3.10.6 Alternate Work Sites a. Determine alternate work sites allowed for use by employees. b. Employ the following security requirements at alternate work sites: [Assignment: organization- defined security requirements] (03.10.06.b). Related Controls: PE-17 ODP Values ODP Identifier ODP Assignment Text ODP Value 03.10.06.b [Assignment: organization-defined security requirements] adequate security, comparable to organizational security requirements at the primary work site where practical, documented in policy, and covered by training Risk Assessment 3.11.1 Risk Assessment a. Assess the risk (including supply chain risk) of unauthorized disclosure resulting from the processing, storage, or transmission of CUI. b. Update risk assessments [Assignment:

NIST SP 800-53 Rev 5 control PE-15.1 (Automation Support) summarizes the requirements for this control: PE-15.1 · Automation Support Control Statement: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Automated mechanisms include notification systems, water detection sensors, and alarms.

NIST SP 800-53 Rev 5 control PE-11.2 (Alternate Power Supply · Self-contained) summarizes the requirements for this control: PE-11.2 · Alternate Power Supply · Self-contained Control Statement: Provide an alternate power supply for the system that is activated [organization-defined parameter] and that is: (a) Self-contained; (b) Not reliant on external power generation; and (c) Capable of maintaining [organization-defined parameter] in the event of an extended loss of the primary power source. Supplemental Guidance: Provide an alternate power supply for the system that is activated [organization-defined parameter] a

Under the FedRAMP Moderate baseline, PE-15 (PE-15 Water Damage Protection): FedRAMP Moderate Baseline · PE-15 · Water Damage Protection Control Statement: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. Supplemental Guidance: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. The provision of water damage protection prima

Under the FedRAMP Moderate baseline, PE-2 (PE-2 Physical Access Authorizations): FedRAMP Moderate Baseline · PE-2 · Physical Access Authorizations Control Statement: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access list when access is no longer required. Supplemental Guidance: a. Develo

NIST SP 800-53 Rev 5 control PE-15.1 (Automation Support) is defined as follows: PE-15.1 · Automation Support Control Statement: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Supplemental Guidance: Detect the presence of water near the system and alert [organization-defined parameter] using [organization-defined parameter]. Automated mechanisms include notification systems, water detection sensors, and alarms.

NIST SP 800-53 Rev 5 control PE-8.3 (Limit Personally Identifiable Information Elements) summarizes the requirements for this control: PE-8.3 · Limit Personally Identifiable Information Elements Control Statement: Limit personally identifiable information contained in visitor access records to the following elements identified in the privacy risk assessment: [organization-defined parameter]. Supplemental Guidance: Limit personally identifiable information contained in visitor access records to the following elements identified in the privacy risk assessment: [organization-defined parameter]. Organizations may have requirement

NIST SP 800-171 Rev 3 control 03.10.08 (03.10.08 Access Control for Transmission) requires organizations to implement the following: 03.10.08 · Access Control for Transmission Supplemental Guidance: Safeguarding measures applied to system distribution and transmission lines prevent accidental damage, disruption, and physical tampering. Such measures may also be necessary to prevent eavesdropping or the modification of unencrypted transmissions. Safeguarding measures used to control physical access to system distribution and transmission lines include disconnected or locked spare jacks, locked wiring closets, cabling protecti For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures, and technical controls address each sub-requirement of 03.10.08.

Under the FedRAMP High baseline, PE-16 (PE-16 Delivery and Removal): FedRAMP High Baseline · PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas and

Under the FedRAMP Low baseline, PE-1 (PE-1 Policy and Procedures): FedRAMP Low Baseline · PE-1 · Policy and Procedures Control Statement: a. Develop, document, and disseminate to [organization-defined parameter]: 1. [organization-defined parameter] physical and environmental protection policy that: 2. Procedures to facilitate the implementation of the physical and environmental protection policy and the associated physical and environmental protection controls; b. Designate an

[organization-defined parameter] to manage the development, documentation, and

NIST SP 800-53 Rev 5 control PE-3.2 (Facility and Systems) is defined as follows: PE-3.2 · Facility and Systems Control Statement: Perform security checks [organization-defined parameter] at the physical perimeter of the facility or system for exfiltration of information or removal of system components. Supplemental Guidance: Perform security checks [organization-defined parameter] at the physical perimeter of the facility or system for exfiltration of information or removal of system components. Organizations determine the extent, frequency, and/or randomness of security checks to adequately mitigate risk associated with exfiltration.

NIST SP 800-53 Rev 5 control PE-20 (Asset Monitoring and Tracking) summarizes the requirements for this control: PE-20 · Asset Monitoring and Tracking Control Statement: Employ [organization-defined parameter] to track and monitor the location and movement of [organization-defined parameter] within [organization-defined parameter]. Supplemental Guidance: Employ [organization-defined parameter] to track and monitor the location and movement of [organization-defined parameter] within [organization-defined parameter]. Asset location technologies can help ensure that critical assets-including vehicles, equip

NIST SP 800-53 Rev 5 control PE-13.4 (Inspections) summarizes the requirements for this control: PE-13.4 · Inspections Control Statement: Ensure that the facility undergoes [organization-defined parameter] fire protection inspections by authorized and qualified inspectors and identified deficiencies are resolved within [organization-defined parameter]. Supplemental Guidance: Ensure that the facility undergoes [organization-defined parameter] fire protection inspections by authorized and qualified inspectors and identified deficiencies are resolved within [organization-defined parameter].

From the NIST SP 800-172 Rev 3 Final Public Draft: ADVERSARY EFFECTS 1450 P reclude (Preempt) REFERENCES Source Control: PE-16 3.11 . Risk Assessment 03.1 1.01E Threat Awareness Program Implement a threat awareness program that includes a cross-organization information-sharing capability for threat intelligence. DISCUSSION Because of the constantly changing and increasing sophistication of adversaries, especially the advanced persistent threat (APT), it may be likely that adversaries can successfully breach or compromise organizational systems. One of the techniques that organizations can use to address this concern is to share threat information. This can include the tactics, techniques, and procedures that organizations have experienced; mitigations that organizations have found to be effective against certain types of th

NIST SP 800-171 Rev 3 control 03.10.06 (03.10.06 Alternate Work Site) requires organizations to implement the following: Control Statement: SR-03.10.06.a Determine alternate work sites allowed for use by employees. SR-03.10.06.b Employ the following security requirements at alternate work sites: [organization-defined parameter]. For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures, and technical controls address each sub-requirement of 03.10.06.

NIST SP 800-53 Rev 5 control PE-9 (Power Equipment and Cabling) is defined as follows: PE-9 · Power Equipment and Cabling Control Statement: Protect power equipment and power cabling for the system from damage and destruction. Supplemental Guidance: Protect power equipment and power cabling for the system from damage and destruction. Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation. Types of power equipment and cabling include internal cabling and uninterruptable power sources in offices or data centers, gen

NIST SP 800-53 Rev 5 control PE-2 (Physical Access Authorizations): PE-2 · Physical Access Authorizations Control Statement: a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides; b. Issue authorization credentials for facility access; c. Review the access list detailing authorized facility access by individuals [organization-defined parameter] ; and d. Remove individuals from the facility access l

03.10.08 · Access Control for Transmission Supplemental Guidance: Safeguarding measures applied to system distribution and transmission lines prevent accidental damage, disruption, and physical tampering. Such measures may also be necessary to prevent eavesdropping or the modification of

unencrypted transmissions. Safeguarding measures used to control physical access to system distribution and transmission lines include disconnected or locked spare jacks, locked wiring closets, cabling protection with conduit or cable trays, and wiretapping sensors. Assessment Objectives: - physical access to system distribution and transmission lines within organizational facilities is controlled.

Under the FedRAMP Low baseline, PE-6 (PE-6 Monitoring Physical Access): FedRAMP Low Baseline · PE-6 · Monitoring Physical Access Control Statement: a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents; b. Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter] ; and c. Coordinate results of reviews and investigations with the organizational incident response capability. Supplemental Guidance: a. Monitor physical access to the facility wh

NIST SP 800-171 Rev 3 control 03.10.08 (03.10.08 Access Control for Transmission) is assessed by examining the following: Assessment Objectives: - physical access to system distribution and transmission lines within organizational facilities is controlled.

NIST SP 800-171 Rev 3 control 03.10.02 (03.10.02 Monitoring Physical Access) is assessed by examining the following: Assessment Objectives: - physical access to the facility where the system resides is monitored to detect physical security incidents. - physical security incidents are responded to. - physical access logs are reviewed [organization-defined parameter] . - physical access logs are reviewed upon occurrence of [organization-defined parameter].

Under the FedRAMP Low baseline, PE-16 (PE-16 Delivery and Removal): FedRAMP Low Baseline · PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas and

NIST SP 800-171 Rev 3 control 03.10.02 (03.10.02 Monitoring Physical Access) requires organizations to implement the following: Control Statement: SR-03.10.02.a Monitor physical access to the facility where the system resides to detect and respond to physical security incidents. SR-03.10.02.b Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter]. For practical implementation, organizations should document their approach in their System Security Plan (SSP) and ensure that policies, procedures, and technical controls address each sub-requirement of 03.10.02.

NIST SP 800-53 Rev 5 control PE-19.1 (National Emissions Policies and Procedures) summarizes the requirements for this control: PE-19.1 · National Emissions Policies and Procedures Control Statement: Protect system components, associated data communications, and networks in accordance with national Emissions Security policies and procedures based on the security category or classification of the information. Supplemental Guidance: Protect system components, associated data communications, and networks in accordance with national Emissions Security policies and procedures based on the security category or classification

Under the FedRAMP Moderate baseline, PE-9 (PE-9 Power Equipment and Cabling): FedRAMP Moderate Baseline · PE-9 · Power Equipment and Cabling Control Statement: Protect power equipment and power cabling for the system from damage and destruction. Supplemental Guidance: Protect power equipment and power cabling for the system from damage and destruction. Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation.

NIST SP 800-53 Rev 5 control PE-12 (Emergency Lighting) summarizes the requirements for this control: PE-12 · Emergency Lighting Control Statement: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. Supplemental Guidance: Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility. The provision of emergency ligh

Under the FedRAMP Moderate baseline, PE-16 (PE-16 Delivery and Removal): FedRAMP Moderate Baseline · PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas

Under the FedRAMP High baseline, PE-17 (PE-17 Alternate Work Site): FedRAMP High Baseline · PE-17 · Alternate Work Site Control Statement: a. Determine and document the [organization-defined parameter] allowed for use by employees; b. Employ the following controls at alternate work sites: [organization-defined parameter]; c. Assess the effectiveness of controls at alternate work sites; and d. Provide a means for employees to communicate with information security and privacy personnel in case of incidents. Supplemental Guidance: a. Determine and document the [o

NIST SP 800-53 Rev 5 control PE-15 (Water Damage Protection) summarizes the requirements for this control: PE-15 · Water Damage Protection Control Statement: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. Supplemental Guidance: Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel. The provision of water damage protection primarily applies to organization

PE-16 · Delivery and Removal Control Statement: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Supplemental Guidance: a. Authorize and control [organization-defined parameter] entering and exiting the facility; and b. Maintain records of the system components. Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas and isolating the areas from the system and media libraries.

Under the FedRAMP High baseline, PE-6 (PE-6 Monitoring Physical Access): FedRAMP High Baseline · PE-6 · Monitoring Physical Access Control Statement: a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents; b. Review physical access logs [organization-defined parameter] and upon occurrence of [organization-defined parameter] ; and c. Coordinate results of reviews and investigations with the organizational incident response capability. Supplemental Guidance: a. Monitor physical access to the facility w

NIST SP 800-171 Rev 3 control 03.10.08 (03.10.08 Access Control for Transmission) is assessed by examining the following: Assessment Objectives: - physical access to system distribution and transmission lines within organizational facilities is controlled.

NIST SP 800-171 Rev 3 control 03.10.08 (03.10.08 Access Control for Transmission) falls under the Physical Protection family. 03.10.08 · Access Control for Transmission Supplemental Guidance: Safeguarding measures applied to system distribution and transmission lines prevent accidental damage, disruption, and physical tampering. Such measures may also be necessary to prevent eavesdropping or the modification of unencrypted transmissions. Safeguarding measures used to control physical access to system distribution and transmission lines include disconnected or locked spare jacks, locked wiring closets, cabling protecti Supplemental Guidance: Safeguarding measures applied to system distribution and transmission lines prevent accidental damage, disruption, and physical tampering. Such measures may also be necessary to prevent eavesdropping or the modification of unencrypted transmissions. Safeguarding measures used to control physical access to system distribution and transmission lines include disconnected or locked spare jacks, locked wiring closets, cabling protection with conduit or cable trays, and wiretapp

NIST SP 800-53 Rev 5 control PE-11 (Emergency Power) is assessed by examining, interviewing, and testing: PE-11 · Emergency Power Control Statement: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. Supplemental Guidance: Provide an uninterruptible power supply to facilitate [organization-defined parameter] in the event of a primary power source loss. An uninterruptible power supply (UPS) is an electrical system or mechanism that provides emergency power when there is a failure of the main

power source. A UPS is typi

NIST SP 800-53 Rev 5 control PE-3.5 (Tamper Protection) summarizes the requirements for this control: PE-3.5 · Tamper Protection Control Statement: Employ [organization-defined parameter] to [organization-defined parameter] physical tampering or alteration of [organization-defined parameter] within the system. Supplemental Guidance: Employ [organization-defined parameter] to [organization-defined parameter] physical tampering or alteration of [organization-defined parameter] within the system. Organizations can implement tamper detection and prevention at selected hardware components or implem

NIST SP 800-53 Rev 5 control PE-18 (Location of System Components) summarizes the requirements for this control: PE-18 · Location of System Components Control Statement: Position system components within the facility to minimize potential damage from [organization-defined parameter] and to minimize the opportunity for unauthorized access. Supplemental Guidance: Position system components within the facility to minimize potential damage from [organization-defined parameter] and to minimize the opportunity for unauthorized access. Physical and environmental hazards include floods, fires, tornadoes, earthqu